

OSTROV ANDREYA, Russian Federation

WMO: 213010

Lat: 76.779N

Lon: 110.787E

Elev: 6

StdP: 101.25

Time Zone: 7.00 (E07)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.2	-42.1	-46.1	0.0	-43.9	-44.3	0.0	-42.0	23.9	-21.8	21.2	-22.2	3.2	210	1.221

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	12.6	10.6	9.4	7.9	7.2	6.1	10.8	12.5	8.1	9.3	6.2	7.1	4.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	7.4	12.1	7.0	6.2	9.0	5.2	5.5	7.0	31.3	12.5	25.1	9.4	21.0	7.2	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.4	12.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-45.7	15.0	2.3	3.2	-47.3	17.3	-48.7	19.2	-50.0	21.0	-51.7	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-16.0	-31.1	-31.2	-28.8	-22.9	-10.5	-1.3	3.1	2.3	-2.8	-15.3	-25.9	-28.2		
	DBStd	14.31	7.89	7.13	7.28	6.37	4.55	3.07	3.17	3.24	4.41	7.60	6.18	6.74		
	HDD10.0	9477	1274	1153	1204	987	634	340	217	240	383	783	1077	1184		
	HDD18.3	12514	1532	1387	1462	1237	892	590	472	497	633	1042	1327	1443		
	CDD10.0	5	0	0	0	0	0	0	3	2	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	5.7	5.6	5.6	5.8	6.0	5.2	5.7	5.3	5.7	5.6	6.0	5.5	6.2		
(15) Precipitation	PrecAvg	269	14	13	12	16	22	20	31	37	37	33	21	15		
	PrecMax	411	52	33	22	46	44	40	57	77	83	64	39	40		
	PrecMin	147	1	1	1	0	3	1	6	11	12	13	7	3		
	PrecStd	71	11	9	7	10	9	10	13	17	17	11	9	9		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.7	-10.5	-6.7	-2.7	1.4	9.4	17.6	17.6	9.9	0.8	-6.4	-8.4		
		MCWB	-7.9	-10.8	-7.2	-3.1	0.9	7.9	15.0	14.5	8.4	-0.4	-6.8	-9.1		
	2%	DB	-12.7	-14.7	-10.6	-7.6	-0.6	5.3	13.5	12.6	6.6	-0.9	-11.4	-13.0		
		MCWB	-13.1	-15.1	-11.2	-8.2	-1.4	4.1	11.4	10.6	5.5	-1.5	-11.7	-13.5		
	5%	DB	-16.7	-16.8	-14.7	-10.3	-2.6	2.9	10.1	9.4	3.8	-2.7	-14.7	-15.7		
		MCWB	-17.0	-17.1	-15.0	-10.8	-3.3	1.8	8.5	7.9	2.8	-3.2	-15.0	-16.0		
	10%	DB	-19.2	-20.2	-18.1	-13.0	-4.0	1.7	7.5	7.3	2.4	-4.5	-17.0	-18.4		
		MCWB	-19.4	-20.4	-18.3	-13.3	-4.6	0.8	6.3	6.2	1.7	-5.1	-17.3	-18.6		
	0.4%	WB	-8.3	-10.7	-7.4	-3.1	0.8	8.0	15.7	14.7	8.3	-0.2	-6.6	-9.0		
		MCDB	-7.8	-10.5	-7.0	-2.4	1.3	9.4	17.5	17.4	9.6	0.5	-6.3	-8.7		
	2%	WB	-13.0	-14.9	-11.4	-8.1	-1.4	4.0	11.4	10.8	5.5	-1.5	-11.8	-13.3		
		MCDB	-12.6	-14.6	-10.9	-7.6	-0.7	5.3	13.3	12.4	6.7	-0.9	-11.5	-13.1		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	-16.9	-17.1	-15.0	-10.7	-3.3	2.0	8.7	8.1	2.9	-3.2	-15.0	-16.0		
		MCDB	-16.8	-16.8	-14.7	-10.3	-2.6	2.7	10.0	9.2	3.6	-2.7	-14.7	-15.7		
	10%	WB	-19.4	-20.4	-18.4	-13.3	-4.6	1.0	6.5	6.3	1.7	-5.0	-17.2	-18.6		
		MCDB	-19.2	-20.2	-18.1	-13.0	-4.0	1.5	7.5	7.3	2.3	-4.5	-17.0	-18.3		
	5% DB	MDBR	6.1	6.5	7.1	7.8	5.4	2.9	4.3	4.1	3.8	6.2	6.6	6.1		
		MCDBR	7.7	10.7	9.8	8.5	5.9	4.9	10.2	9.5	5.3	6.0	9.9	7.8		
(36) Mean Daily Temperature Range	5% WB	MCWBR	7.5	10.5	9.8	8.4	5.5	3.7	8.6	7.8	4.4	5.9	9.8	7.7		
		MCDBR	7.8	10.4	10.4	8.4	5.8	4.5	10.3	9.3	4.9	6.3	10.0	7.8		
	5% WB	MCWBR	7.5	10.2	10.2	8.3	5.4	3.6	8.8	7.8	4.2	6.2	9.8	7.6		
		MCDBR	7.5	10.2	10.2	8.3	5.4	3.6	8.8	7.8	4.2	6.2	9.8	7.6		
	Clear-Sky Solar Irradiance	taub	N/A	0.171	0.258	0.296	0.321	0.318	0.354	0.334	0.273	0.186	N/A	N/A		
		taud	N/A	1.874	1.985	2.047	2.075	2.307	2.352	2.417	2.342	1.949	N/A	N/A		
		Ebn at Noon	0	128	570	751	800	834	776	731	610	87	0	0		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.00	0.05	1.03	3.38	5.08	5.13	4.13	2.53	0.96	0.15	0.00	0.00		
		RadStd	0.00	0.01	0.10	0.28	0.44	0.61	0.34	0.27	0.10	0.02	0.00	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page